

Does a Music Transformer Know What Key It Is In?

Causal Interventions on an Emergent Tonal World Model

Anonymous ICASSP 2027 submission

Abstract—Music generation models produce convincingly tonal music. Whether they chain surface statistics or maintain an internal representation of musical key that generation *causally uses* has not been tested. We test it directly, under a pre-registered protocol with frozen decision rules, on Transformers trained on symbolic music whose vocabulary holds absolute pitch and time only—no key, chord, or degree tokens. Replacing the key component of the residual stream at a single middle layer makes the model continue in the injected key at $5.6\times$ the rate of rank- and norm-matched random edits (guarded TKR 0.378 vs. 0.068; significant for 12/12 targets after Holm correction, replicated across seeds), inside a musicality budget frozen before any edit ran, reaching 58% of the ceiling set by transposing the prompt itself. Two controls say what this requires. Across a $50\times$ capacity range next-token accuracy is flat, yet at 1.6M parameters the key is not decodable beyond the input surface at all and the same edit moves the key *more* often than at 25M while wrecking the music 90% of the time: capacity does not buy the key, it buys a key that can be moved without breaking everything else. Finally we leave our own models: on real Bach chorales a *public* checkpoint trained on real music—architecturally identical to one of ours—carries a key state that beats the pitch surface where ours does not, and editing that state redirects the music it composes at $5.7\times$ the matched control (11/12 targets, 100% within the frozen musicality budget). It also resists: the keys the edit cannot reach are the keys real music rarely uses. Editing a music Transformer’s internal key state changes what key it composes in—in a model people actually use, not only in ours.

Index Terms—music generation, world models, interpretability, causal intervention, tonality

I. INTRODUCTION

Systems that compose music write convincingly tonal pieces: they settle into a home key, build tension away from it, and return. How they do so is unexamined. Analyses report that musical attributes can be *read out* of a model’s activations by trained classifiers [6]. Reading is not using: a signal can be decodable from a system’s state without the system consulting it when it acts [10]. No study has tested whether a music model *uses* an internal key representation—whether intervening on it, with the input untouched, changes the key of the music it goes on to generate.

Sequence models can build causal models of the latent state behind their data. A model trained only to predict

Othello moves tracks the board those moves imply, and editing that internal board changes which moves it treats as legal [1], [2]; analogous causal evidence exists for chess, mazes, and RL environments [3], [4], [5]. In music, evidence has stopped at readability [6], and steering work controls attributes without testing a state variable [7]. We occupy the empty cell: causal, state-level evidence in music.

Key is an unusually good state variable for this. It is defined *externally* to the model—by centuries of theory and by psychological measurement [9]—so the probed state is not an artifact of our labeling; it must be *computed* by integrating pitch evidence over time, unlike metric position, which time tokens hand over; and diatonic membership gives a natural analogue of Othello’s legal-move rate, so success is measurable without a learned judge.

Contributions: (1) a controlled testbed—leak-free tokenizer, functional-harmony corpus with exact per-token key labels, pre-registered protocol—released as a benchmark; (2) a causal result: single-layer subspace edits redirect the composed key for 12/12 targets inside a frozen musicality budget; (3) two controls that say what a world model requires: a capacity sweep separating the state’s presence, its legibility, and its manipulability; and the same probe AND the same intervention carried to a public model trained on real music, which reads its key state and uses it.

II. METHOD

A. Data, models, protocol

Ground truth must be exact, not estimated, so training data are synthetic. **D-SYN**: a functional-harmony state machine ($T \rightarrow S \rightarrow D \rightarrow T$ with substitutions, SATB voicing, optional diatonic melody) generates 200k/10k/10k train/val/test pieces of 278–508 tokens with 0–3 modulations each (pivot, direct, or sequential; fifths-distance stratified), giving every token a ground-truth key label $\kappa(t)$ (tonic $\times$ mode, 24 classes). **Tokenizer**: the vocabulary (124 types) holds only BAR, POS, absolute-MIDI PITCH, and DUR; a unit test enforces the absence of key/chord/degree tokens, so key must be computed from the notes. **Models**: M-CTRL is a GPT-2 style decoder (8 layers, 8 heads, $d=512$, ctx 512, ~ 25 M params), trained under two regimes—with (R-Aug) and without (R-NoAug) uniform 12-fold transposition augmentation— $\times 3$ seeds. M-REF shares the architecture but a disjoint seed *and* data split,

and serves only as the musicality guard, avoiding circular evaluation. All models pass a pre-frozen quality gate (val. next-token top-1 0.878–0.881 vs. a 0.434 constant-predictor rate). Decision rules, thresholds and controls were frozen before any experiment ran; deviations are logged and negative results reported.

B. Probing (Phase A)

Linear probes map residual-stream activations $h_\ell(t)$ to $\kappa(t)$ with sequence-level splits, against three control families: **C1**, a control task [10] permuting key identities per sequence; **C2**, an untrained model; **C3**, input-side baselines at the same positions (pitch-class-histogram logistic regression and Krumhansl–Schmuckler (KS) template matching, windows $W \in \{8, 16, 32, 64\}$). **DR-H1**: the probe supports H1 iff its selectivity-corrected F_1 exceeds the best C3 baseline with a sequence-level BCa 95% CI excluding zero in some layer.

C. Intervention (Phase B)

From bar 9 of a key-stable 8-bar prompt, a sustained edit replaces the key component at one layer,

$$h \leftarrow h - P_V h + P_V \mu_{\kappa^*}, \quad (1)$$

with V an orthonormal key subspace, $P_V = VV^\top$, and μ_{κ^*} the class-conditional mean of the donor key κ^* . Subspaces: **V-PROBE** (probe row space, rank 24), **V-MEAN** (class-mean span), **V-DAS** (interchange-trained, ranks 8/24). Controls: **K1** rank/norm-matched random subspace; **K2** sham edit, required bit-identical to the clean run (passed, 100/100); **K3** another layer’s subspace; **K4** the behavioral ceiling—transposing the prompt itself. Continuations (16 bars; 12 major targets $\times$ 100 prompts $\times$ 8 layers; sampling noise paired with a clean twin) are scored by KS key estimation (TKR: the continuation’s estimated key equals κ^*) and by IKR (fraction of continuation pitches diatonic to a key). The musicality budget $\delta_{\text{PPL}}=0.613$ —the 90th percentile of the M-REF perplexity rise across 7,989 *natural* modulations in validation data—was fixed *before* any edit ran; edits exceeding it do not count as successes. **DR-H3** requires $\text{TKR}(\text{edit}) > \text{TKR}(\text{K1})$ with Holm-corrected Wilcoxon $p < .05$ in $\geq 8/12$ targets at the best in-budget condition.

III. RESULTS

Intervention numbers are from the R-Aug seed-0 sweep; probing holds for all six M-CTRL models, and the seed-1 sweep replicates every Phase-B verdict.

A. The key is decodable beyond the input surface

The key is *computed*, not transcribed: at L4 the linear probe reaches macro- F_1 0.927 against 0.790 for the best input baseline (PC-hist LR, $W=64$) and a 0.032 selectivity floor, so the corrected margin is +0.105 (BCa 95% CI [0.091, 0.114]). The margin excludes zero in layers 1–7 of every trained model and fails only at L0 (−0.20): **DR-H1**

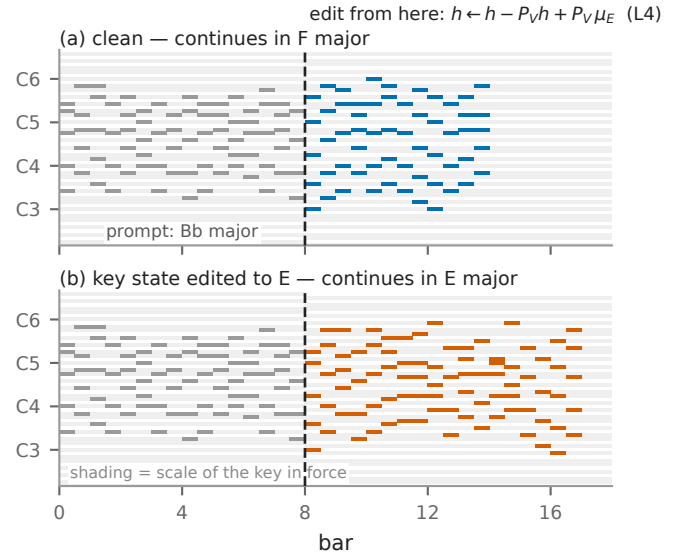


Fig. 1. The claim in one example (real model output, fixed sampling seed). The same 8-bar Bb-major prompt, continued (a) clean and (b) with the key component of layer-4 activations replaced by the E-major class mean from bar 9 on. **The input does not change**; the continuation’s estimated key follows the injected internal state. Shading marks the scale of the key in force.

is supported 6/6. An untrained model scores 0.243. The internal state is worth most where the surface is worst: in the top quartile of KS-entropy ambiguity, probe accuracy is 0.936 vs. 0.795 for the baseline.

B. Fifths geometry; equivariant but not a group

The representation carries music-theoretic structure. Projected to two principal components, the twelve major-tonic probe directions order themselves as C–G–D–A–E–B–F♯... (Fig. 2): circle-of-fifths neighbour agreement 1.00 vs. 0.00 for the chromatic ordering, at every probed layer (exploratory). Transposition acts as a generalizing linear map—Procrustes R_k fit on half the sequences aligns held-out activations with 0.05–0.11 relative error—but the family is not cyclic ($\varepsilon_{\text{cyc}} \approx 0.98$, vs. ≈ 1.41 for random orthogonal maps), and our pre-registered comparison is *negative*: augmentation does not improve cyclicity (3 seeds per regime; 0.983 vs. 0.981; one-sided $p = 0.83$), so **DR-H2b** is not supported.

C. Editing the key state changes the composed key

At the best in-budget condition (V-PROBE, L4), $\text{TKR}(\text{edit}) - \text{TKR}(\text{K1}) > 0$ holds for 12/12 targets (Holm-corrected $p < 10^{-4}$; rank-biserial $r = 0.64$ –1.00) against a pre-registered bar of $\geq 8/12$: **DR-H3 is supported** (Fig. 3). Pitch content triangulates the same conclusion without the estimator: $\text{IKR}_{\kappa^*}=0.937$ vs. $\text{IKR}_{\kappa_{\text{src}}}=0.646$ after editing. Success falls gently with fifths distance (0.395 at distance 1 to 0.280 at the tritone), while random edits collapse to zero once the identity target is excluded. The edit reaches 58% of the behavioral ceiling (K4): the state is

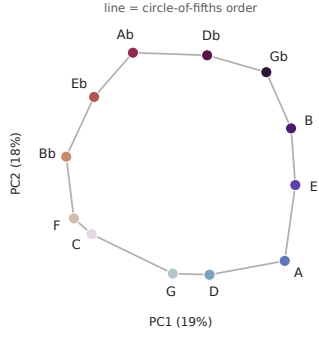


Fig. 2. The twelve major-tonic probe directions (L4, first two PCs) self-organize into a circle of fifths. No key, chord, or degree token exists in the vocabulary.

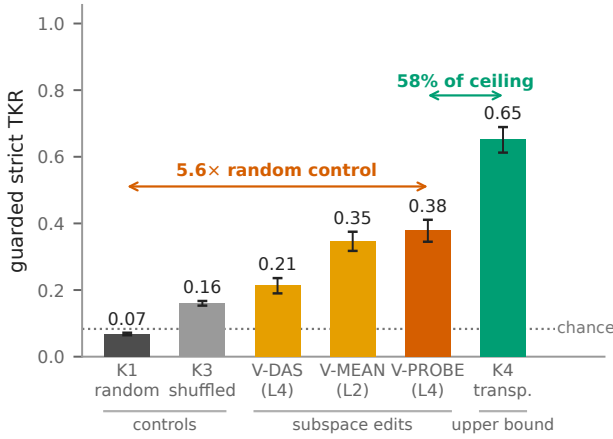


Fig. 3. Guarded strict TKR by condition: the continuation lands exactly on the injected key *and* stays inside the frozen musicality budget. Chance = $1/12$. Bars: prompt-level bootstrap 95% CI.

readable in a rank-24 subspace of one layer, yet a single-layer edit realizes only part of the behavioral effect—an *actionability gap*. Two subsidiary findings: the *optimized* subspace (V-DAS, 0.213) underperforms the probe’s read-out directions, and V-MEAN edits destroy quality at deep layers (guard pass $\leq 5\%$)—the guard is load-bearing, as Sec. III-E makes decisive.

D. Readability is diffuse; causal efficacy is sharp

The edit-minus-control effect is single-peaked across layers (0.03, 0.04, 0.19, 0.26, **0.30**, 0.26, 0.23, 0.23 for L0–L7); the best-vs. -median bootstrap CI [0.038, 0.109] excludes zero (DR-H5 supported). The two profiles in Fig. 4 are *not* the same shape, and that is the finding. The probe saturates across L2–L6 (macro- F_1 0.920–0.929, all within 0.01 of maximum): the key is legible from most of the network. Yet a single-layer edit moves the output only near L4, and barely at all at L0–L1. *Where a state is legible is a weak guide to where it is used*—a caution for work that localizes a computation by probing accuracy alone.

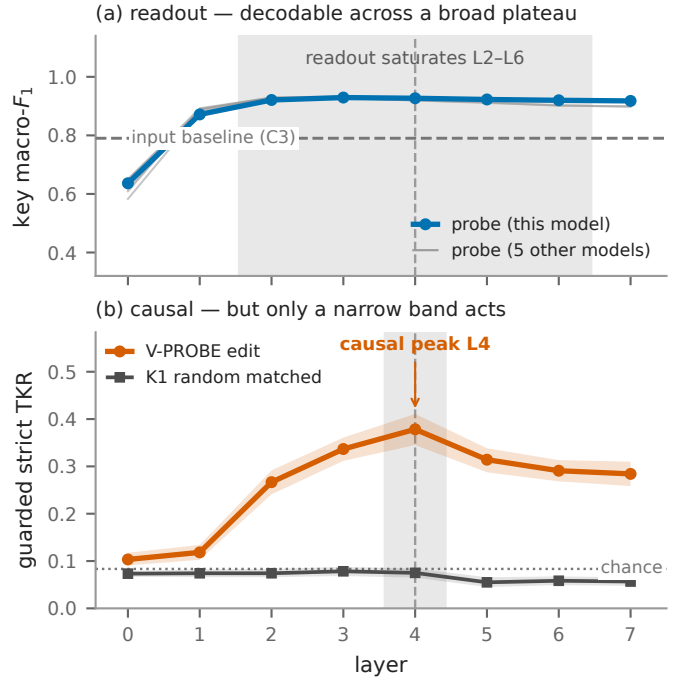


Fig. 4. (a) Probe macro- F_1 per layer, all six M-CTRL models (thin lines; R-Aug seed 0 highlighted): the key is legible across a broad plateau. (b) Causal effect: guarded TKR of the V-PROBE edit vs. the K1 matched random control (bands: prompt-level bootstrap 95% CI). Only a narrow band around L4 moves the output.

E. What a world model actually requires

Training the same data and protocol at 1.6M, 6M, 25M and 85M parameters dissociates three things the term “world model” conflates (Table I, top). *Behaviour* is flat: next-token top-1 is 0.868–0.879 across a $50\times$ range. *Legibility* is not: at 1.6M the probe peaks *below* the input baseline and DR-H1 is unsupported in both seeds—a behaviourally equivalent model with no key state, the surface-statistics account made flesh, which the pre-registered test correctly refuses to credit. *Manipulability* is a third thing again: at 1.6M the edit moves the key *more* often than at 25M in raw terms (0.55 vs. 0.43) and survives the musicality guard only 10% of the time (vs. 80%)—with two layers, “editing the key subspace” is a blunt overwrite, not the movement of a separable state. Reporting guarded TKR alone hides this; reporting raw TKR alone credits a surface model with a world model. The guard is what tells them apart. Nor does manipulability simply keep improving: it peaks near 25M and falls to 0.18 at 85M, and a full twelve-layer sweep confirms this is not a layer-choice artifact (the 85M causal peak, L6 at 0.206, is indistinguishable from the L3 we had used, 0.202).

F. A public model trained on real music: it reads, and it uses

Everything above is our own model on our own data. We close by leaving both. On 300 Bach chorales whose

local key at every note comes from published Roman-numeral analyses,¹ our 85M model’s key subspace transfers (F_1 0.534 vs. a 0.037 selectivity floor) but does *not* beat the pitch surface: the corrected margin, +0.034 (CI [−0.021, 0.103]), does not exclude zero.

The Anticipatory Music Transformer [8] (Apache-2.0; Lakh MIDI, MetaMIDI, FMA, 450k commercial records) is, at its **small** size, *architecturally identical* to that model—12 layers, $d=768$ —and its vocabulary, (arrival-time, duration, note) triples, is leak-free exactly as ours is. Probed on the same chorales, at the same positions, with the same controls and rule (the two C3 baselines land at 0.451 and 0.454, confirming the setups match), it reaches F_1 0.690 with a corrected margin of +0.196 (CI [0.132, 0.260]): **DR-H1 supported**. With architecture, corpus, protocol and rule fixed, *the training data is the single free variable*, and it decides whether a key state is worth more than the notes themselves.

And that state is used. We ran the same intervention on it: the probe’s row space as the subspace, a *different* public checkpoint as the guard’s reference (medium refereeing small—different weights, not circular; $\delta_{\text{PPL}}=0.849$ nats frozen from 1,180 natural modulations in the chorales), the edit layer searched on 20 chorale prefixes and judged on 60 it never saw, sham edit bit-identical (20/20). **DR-H3 is supported**: at L8 the edit lands the continuation on the injected key at $5.7\times$ the matched random control (0.365 vs. 0.064), significant in 11/12 targets, with IKR crossing (0.824 to the injected key vs. 0.785 to the prompt’s) and—unlike our own model’s 87%—100% **of edits inside the frozen musicality budget**. A key state learned from real music is *more* cleanly separable, not less.

Two things only a real model could show. The causal peak (L8) is again not the probe peak (L10), so Sec. ??’s dissociation is not an artifact of synthetic data. And the model *resists*: the one target it cannot reach is F $\sharp$ major (TKR 0.000), and across all twelve targets success is ordered almost perfectly by how common that tonic is in real music (Spearman +0.90). A model trained on real music can be steered into keys it knows and resists keys its training distribution barely contains—a prior our uniformly-keyed synthetic corpus could not have revealed. One caution earned the hard way: the probe must be read where the model is about to *choose* a pitch; read at the note itself, after the pitch is committed, the same model’s profile decays with depth and the key appears to vanish. Where one probes is not a free parameter.

IV. DISCUSSION AND LIMITATIONS

The evidence supports the world-model reading within this training condition: a Transformer trained on notes alone maintains a mid-layer key variable with music-theoretic geometry, and generation consults it. The gap to

¹Scores: **bach-370-chorales** (CC BY-NC-SA); analyses: *When in Rome* (CC BY-SA). Neither is redistributed.

TABLE I
WHAT A WORLD MODEL REQUIRES. **Top**: SAME DATA AND PROTOCOL, FOUR CAPACITIES — BEHAVIOUR IS FLAT WHILE LEGIBILITY AND MANIPULABILITY ARE NOT. **Bottom**: SAME ARCHITECTURE (12 LAYERS, $d=768$), SAME CHORALES, SAME PROTOCOL; ONLY THE TRAINING DATA DIFFERS. THE PUBLIC MODEL BOTH READS THE KEY AND USES IT.

Model	top-1	probe	DR-H1	guard
<i>capacity (D-SYN, our models)</i>				
1.6M	0.870	below C3	no	10%
6M	0.878	+0.05	yes	60%
25M	0.879	+0.10	yes	80%
85M	0.879	+0.10	yes	71%
<i>real music, matched architecture (Bach chorales)</i>				
ours, synthetic-trained	—	+0.034	no	—
public, real-trained [8]	—	+0.196	yes	—

the behavioral ceiling, the failure of edits at L0–L1, and the partial effect of layer-shuffled subspaces ($K3 = 0.160$, between K1 and the true edit) point to a distributed, redundantly re-estimated state rather than a single canonical register—a contrast with Othello-GPT’s near-complete editability that we attribute to music’s continual re-grounding of tonality in the immediately preceding notes.

Bounds. *V-DAS below V-PROBE*: the optimized subspace may be undertrained relative to its search space; we report it as observed and release its configuration. *Estimator*: the KS judge confuses fifths-neighbours, which strict/tolerant reporting brackets and IKR triangulates. *Capacity*: the sweep varies depth and width together, locating a threshold, not its cause. *Causality on real models*: Sec. III-E shows the *readout* generalizes to a public real-trained model; we have not intervened on it, so it inherits no causal claim—that, and whether edited continuations modulate *musically* (pivot-like transitions vs. jumps), are the journal version’s questions. Supplementary material carries the ambiguity strata, the 12×24 specificity matrix, the fifths-distance curve, and per-seed variants of every figure. All training data are synthetic; the evaluation corpora are used under their stated licenses.

V. CONCLUSION

We gave pre-registered causal evidence that a music Transformer tracks musical key as an internal state variable that generation consults, and quantified how far a single-layer edit goes toward full behavioral control. Two controls say what this requires: capacity supplies not the key but a key that can be moved without breaking the music, and it is the training data—not the architecture—that decides whether such a state beats the notes themselves. The testbed (exact state labels, leak-free vocabulary, frozen decision rules) is released as a benchmark. Music offers state variables that theory defines and ears can verify; our next question is whether models also internalize how keys are *supposed* to change.

Reproducibility. Anonymized code, data generator, pre-registration, and an audio demo page (clean vs. edited continuations): <https://anonymous.4open.science/r/TBD>. **[TBD (run required): create before submission.]**

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- [5] **[TBD (run required): RL world-model probing reference; finalize]**.
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